

JAMES RICHARD PATTERSON

Cambridge, United Kingdom

Email: james.patterson@gmail.com

Phone: +44 7921 445678

LinkedIn: linkedin.com/in/jamespatterson-engineer

PROFESSIONAL SUMMARY

Results-driven Principal Engineer with a proven track record of spearheading mission-critical initiatives across dynamic organisations. Highly motivated to leverage cutting-edge technologies and synergistically champion cross-functional collaboration to deliver actionable insights and accelerate organisational objectives. Passionate about data-driven decision making and passionate about delivering value through innovative solutions. Committed to architecting scalable systems that enable proven organisational growth and transformation.

TECHNICAL SKILLS

Architecture and Design: Microservices Architecture, System Design, Cloud Architecture, Distributed Systems, Enterprise Architecture Patterns

Cloud Platforms: AWS, Microsoft Azure, Google Cloud Platform, Infrastructure as Code

Container Orchestration: Kubernetes, Docker, Docker Swarm, Container Security

Infrastructure and Automation: Terraform, Ansible, CloudFormation, ArgoCD, CI/CD Pipelines, Jenkins, GitLab CI

Programming Languages: Java, Python, Go, TypeScript, SQL, Bash

Databases: PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, Elasticsearch

Messaging and Streaming: Apache Kafka, RabbitMQ, AWS SQS, Apache Spark

Development Practices: Agile, Scrum, DevOps, Test-Driven Development, System Design

PROFESSIONAL EXPERIENCE

Principal Engineer

Velocity Systems Ltd | Cambridge, United Kingdom | velocity-systems.io

October 2022 - Present

Architected and led the design of a large-scale microservices platform supporting 500+ API endpoints serving 2 million daily requests, resulting in 40% improvement in system throughput and 25% reduction in infrastructure costs

Established engineering excellence standards across the organisation through creation of comprehensive architectural guidelines, design review processes, and technical documentation that improved code quality metrics by 35%

Mentored and coached a team of 12 engineers across multiple time zones, facilitating professional development and fostering a culture of continuous learning that resulted in three internal promotions

Implemented disaster recovery procedures and achieved 99.99% uptime SLA across all production systems through proactive monitoring, automated incident response, and regular chaos engineering exercises

Senior Principal Engineer

TechCore Innovation Partners | Cambridge, United Kingdom | www.techcore.com

November 2020 - October 2022

Led a strategic cloud migration programme moving 85 monolithic applications to AWS, reducing operational overhead by 45% and enabling the business to scale to 10x user capacity without additional staffing

Designed and implemented a comprehensive CI/CD pipeline utilising Jenkins and containerised environments that reduced deployment time from 2 hours to 15 minutes and improved release frequency from quarterly to bi-weekly

Established security-first practices including implementation of secrets management, infrastructure encryption, and network segmentation across all systems, eliminating critical security vulnerabilities and achieving SOC 2 Type II compliance

Directed technical recruitment and onboarding of senior engineering talent, building a high-performing platform engineering team of 8 engineers that delivered multiple cross-functional transformation initiatives on time and within budget

Senior Engineer

DataFlow Analytics Ltd | Cambridge, United Kingdom | dataflow-analytics.co

January 2021 - September 2022

Developed and optimised a real-time data streaming platform processing 50 terabytes of data daily using Apache Kafka and Spark, enabling business intelligence teams to generate insights 10x faster than previous batch-based systems

Built comprehensive monitoring and observability solutions using Prometheus, Grafana, and ELK Stack across all production systems, reducing mean time to detection (MTTD) of incidents from 45 minutes to 8 minutes

Spearheaded API standardisation initiative documenting and refactoring 200+ legacy endpoints to conform to modern REST standards and OpenAPI specifications, improving developer experience and reducing integration time by 30%

Mentored three junior engineers in distributed systems design and advanced Java programming, leading to successful delivery of two major system components and direct contribution to team capability development

EDUCATION

Doctor of Philosophy in Computer Science

University of Cambridge, Cambridge, United Kingdom

Completed 2015

Specialisation: Distributed Systems and Fault Tolerance

Master of Science in Computer Science

Imperial College London, London, United Kingdom

Completed 2011

Bachelor of Science in Computer Science with Honours

University of Bristol, Bristol, United Kingdom

Completed 2009